XRPL Validator — Daily Engineering Report · 2026-10-14

Period: 2026-10-14 00:00 to 24:00 US Eastern (2026-10-14 04:00 to 2026-10-15 04:00 UTC); ledgers #107,642,576 to #107,664,777. Every count from the ledger and the validator's logs stops at #107,664,777, closed 03:59:52 UTC. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records; Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data; data.xrpl.org agreement scores. Not available: none.

Status

Impaired (all causes resolved or fixed) — 8 validations of a ledger hash the network did not validate.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 22,158 of 22,202 ledgers; 8 validations did NOT match the network's validated ledger. Unsigned: 44 during planned restarts.
- **Native engine:** 557 engine receipts recorded (557 fixed, 0 explained, 0 without a recorded fix).
- **Changes:** 5 deployments (cycles 172, 173, 174, 175, 176), 40 commits merged into main.
- **Amendment fixCleanup3_4_0** holds majority; activation due ~2026-10-23 09:34 UTC.

Service levels

Measure	Value
Ledgers closed by the network	22,202
Validations issued	22,158
Coverage (validations / ledgers closed)	99.802%
Validations matching the network's validated ledger hash	22,150 of 22,158
Unsigned during planned restarts	44
Unsigned outside maintenance	0
Coverage outside planned maintenance (objective 99.9%)	100.000%
Restarts	5

Measure	Value
Basis: the hash the first peer announced as accepted (before cycle 183)	22,158
External agreement, data.xrpl.org (UTC day 2026-10-14)	0.99788 (43 missed of 20,325)

Restarts:

- 06:43:06 UTC: deployment of cycle 172; validations resumed at #107,645,090 (9 ledgers unsigned between the last validation before the restart and the first after it)
- 14:22:25 UTC: deployment of cycle 173; validations resumed at #107,652,162 (9 ledgers unsigned between the last validation before the restart and the first after it)
- 16:40:22 UTC: deployment of cycle 174; validations resumed at #107,654,288 (10 ledgers unsigned between the last validation before the restart and the first after it)
- 21:21:59 UTC: deployment of cycle 175; validations resumed at #107,658,616 (8 ledgers unsigned between the last validation before the restart and the first after it)
- 03:16:39 UTC: deployment of cycle 176; validations resumed at #107,664,113 (8 ledgers unsigned between the last validation before the restart and the first after it)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Earlier FFI shadow log (libxrpl applied against the validator's mirror; superseded by Step 1): 6 transaction-level differences recorded this day, across 6 ledgers. These were not triaged into receipts.

Ledgers with a receipt in the period (#107,642,576–#107,664,777), by step:

Step	Ledgers with a receipt
Step 1 · native engine	0
Step 2 · records	550
Step 3 · order	7
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	4
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	06:49 UTC	#107,645,179	Step 2 · records	Payment tx 6616BEFDE401: RippleState 1A229E1B (modified on the network): PreviousTxnID ours=null theirs="BD42FEFE10058BC31BE8DE54168…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line,

ID	Time	Ledger(s)	Step	Description	Disposition
					Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-02	06:51 UTC	#107,645,214	Step 2 · records	Payment tx 1C3F97D6E695: DirectoryNode 94479AEC (deleted on the network): FinalFields.PreviousTxnID ours="F6908F4D688CBAF26A8CEE3D036…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-03	06:52 UTC	#107,645,229	Step 2 · records	OfferCreate tx 663BBDCD04C6: DirectoryNode D817686F (deleted on the network): FinalFields.PreviousTxnID ours="CD476AA9133DAD65824BF0818DF…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-04	06:55 UTC	#107,645,271	Step 2 · records	AccountDelete tx 36B39129A459: AccountRoot 4E780827 (deleted on the network): FinalFields.Balance ours="800020" theirs="0"	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-05	06:56 UTC	#107,645,280	Step 2 · records	OfferCreate tx 9AE679AFE3BC: DirectoryNode C09A0331 (deleted on the network): FinalFields.PreviousTxnID ours="21FE1EA2201E01B45E1B1068A20…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-06	06:58 UTC	#107,645,312	Step 2 · records	Payment tx 11AB946350C1: RippleState 14FCA7E2 (modified on the network): PreviousTxnID ours=null theirs="7801169A312500128D9E90C8254…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-07	07:00 UTC	#107,645,345	Step 2 · records	AMMWithdraw tx B261F6DA5A21: RippleState E289EC37 (deleted on the network): FinalFields.Balance ours= {"currency":"03B245BE580EC4F4386A7…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-08	07:01 UTC	#107,645,365	Step 2 · records	OracleSet tx AF7743811680: Oracle 2EFC6BA1 (modified on the network): PreviousFields.PriceDataSeries ours= [{"PriceData":{"AssetPr…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-09	07:01 UTC	#107,645,369	Step 2 · records	OracleSet tx A7F512E8A4E3: Oracle 92CD6679 (modified on the network): PreviousFields.PriceDataSeries ours= [{"PriceData":{"AssetPr…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-10	07:02 UTC	#107,645,373	Step 2 · records	OracleSet tx 62BE59813641: Oracle 3E71E7BD (modified on the network): PreviousFields.PriceDataSeries ours= [{"PriceData":{"AssetPr…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-11	07:02 UTC	#107,645,375	Step 2 · records	OracleSet tx 799B9042E800: Oracle 1850DBB6 (modified on the network): PreviousFields.PriceDataSeries ours=	fixed: Step 2 record fixes (Batch marks, Oracle prices,

ID	Time	Ledger(s)	Step	Description	Disposition
				[{"PriceData":{"AssetPr…	AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-12	07:02 UTC	#107,645,381	Step 2 · records	OracleSet tx EEB606A3017D: Oracle D08E7228 (modified on the network): PreviousFields.PriceDataSeries ours= [{"PriceData":{"AssetPr…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-13	07:02 UTC	#107,645,384	Step 2 · records	OracleSet tx 5FE2C989A358: Oracle B90CC214 (modified on the network): PreviousFields.PriceDataSeries ours= [{"PriceData":{"AssetPr…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-14	07:02 UTC	#107,645,386	Step 2 · records	OracleSet tx 525ED207549F: Oracle 92BE08F2 (modified on the network): PreviousFields.PriceDataSeries ours= [{"PriceData":{"AssetPr…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-15	07:03 UTC	#107,645,388	Step 2 · records	OracleSet tx 27D7744FEE38: Oracle 61AA5830 (modified on the network): PreviousFields.PriceDataSeries ours= [{"PriceData":{"AssetPr…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-16	07:03 UTC	#107,645,390	Step 2 · records	OracleSet tx 760B38D28F6B: Oracle 247316F2 (modified on the network): PreviousFields.PriceDataSeries ours= [{"PriceData":{"AssetPr…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-17	07:03 UTC	#107,645,392	Step 2 · records	OracleSet tx 5B2F767B5AC6: Oracle C99DB849 (modified on the network): PreviousFields.PriceDataSeries ours= [{"PriceData":{"AssetPr…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-18	07:03 UTC	#107,645,394	Step 2 · records	OracleSet tx BFA5A7F8A7FA: Oracle E8232297 (modified on the network): PreviousFields.PriceDataSeries ours= [{"PriceData":{"AssetPr…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-19	07:06 UTC	#107,645,434	Step 2 · records	Payment tx 2E2EFFF0B3A0: RippleState 4C644F01 (modified on the network): PreviousTxnID ours=null theirs="3147F3207BA8A20E978CDA7B804…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-20	07:07 UTC	#107,645,452	Step 2 · records	AMMWithdraw tx B87DE7791DFE: RippleState DED18E33 (deleted on the network): FinalFields.Balance ours= {"currency":"03D65C0022D7ED297D50E…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)

ID	Time	Ledger(s)	Step	Description	Disposition
R-21	07:08 UTC	#107,645,467	Step 2 · records	Payment tx 38EB18B418CA: DirectoryNode F4F861DA (deleted on the network): FinalFields.PreviousTxnID ours="C678C0F2E56E144097D3525DAE4…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-22	07:09 UTC	#107,645,485	Step 2 · records	Payment tx 014D74C3AC46: DirectoryNode E4097E0A (deleted on the network): FinalFields.PreviousTxnID ours="A82E9C39279BE9743AE6EA070C6…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-23	07:13 UTC	#107,645,553	Step 2 · records	Payment tx 7EE57182E69D: RippleState E4808DC8 (modified on the network): PreviousTxnID ours=null theirs="FF7E729EF12B086D9DCD1F4C25E…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-24	07:14 UTC	#107,645,562	Step 2 · records	AMMWithdraw tx 3FAD3C417805: RippleState F490FC6F (deleted on the network): FinalFields.Balance ours= {"currency":"030230B01D2EA24FA6512…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-25	07:16 UTC	#107,645,588	Step 2 · records	AccountDelete tx 58589EDD04A0: AccountRoot 6529E0CD (deleted on the network): FinalFields.Balance ours="800010" theirs="0"	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-26	07:16 UTC	#107,645,592	Step 2 · records	OfferCreate tx CB5A68F9DAEC: DirectoryNode 8FF40377 (deleted on the network): FinalFields.PreviousTxnID ours="290F0FB3ADCE0F5BA006B91F9CD…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-27	07:20 UTC	#107,645,652	Step 2 · records	AccountDelete tx 35CB20120F1B: AccountRoot 9435A836 (deleted on the network): FinalFields.Balance ours="2625172" theirs="0"	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-28	07:24 UTC	#107,645,712	Step 2 · records	OfferCreate tx 725CF18EBC2A: DirectoryNode 10D61123 (deleted on the network): FinalFields.PreviousTxnID ours="447A6AE15B6163D3868C69ED159…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-29	07:24 UTC	#107,645,716	Step 2 · records	Payment tx 7DE61E6AB909: DirectoryNode 99E70624 (deleted on the network): FinalFields.PreviousTxnID ours="4CFC905AA1B934503B26747F751…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-30	07:26 UTC	#107,645,749	Step 2 · records	AccountSet tx A638D82C141D: DirectoryNode FDA39996 (deleted on the network): FinalFields.PreviousTxnID ours="0AB72E6342B9EDCC3A475623DE8…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete,

ID	Time	Ledger(s)	Step	Description	Disposition
					directory links), live since 10-14 10:23 (cycle 173)
R-31	07:31 UTC	#107,645,817	Step 2 · records	OfferCreate tx FF003118FAF6: DirectoryNode ECB3EAF5 (deleted on the network): FinalFields.PreviousTxnID ours="8E2B38AE6434B369049D52E0E11…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-32	07:31 UTC	#107,645,827	Step 2 · records	OracleSet tx 6B3B8BD54E1A: Oracle 785C8834 (modified on the network): PreviousFields.PriceDataSeries ours= [{"PriceData":{"AssetPr…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-33	07:31 UTC	#107,645,829	Step 2 · records	OracleSet tx 72499CD5B421: Oracle 92CD6679 (modified on the network): PreviousFields.PriceDataSeries ours= [{"PriceData":{"AssetPr…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-34	07:32 UTC	#107,645,833	Step 2 · records	OracleSet tx 7B8E5B56201A: Oracle 3E71E7BD (modified on the network): PreviousFields.PriceDataSeries ours= [{"PriceData":{"AssetPr…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-35	07:32 UTC	#107,645,841	Step 2 · records	OracleSet tx 08BCFDCAECE6: Oracle D08E7228 (modified on the network): PreviousFields.PriceDataSeries ours= [{"PriceData":{"AssetPr…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-36	07:33 UTC	#107,645,847	Step 2 · records	OracleSet tx 3FA2475F70FD: Oracle 61AA5830 (modified on the network): PreviousFields.PriceDataSeries ours= [{"PriceData":{"AssetPr…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-37	07:33 UTC	#107,645,849	Step 2 · records	OracleSet tx 6C96FAEEDF08: Oracle 247316F2 (modified on the network): PreviousFields.PriceDataSeries ours= [{"PriceData":{"AssetPr…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-38	07:33 UTC	#107,645,851	Step 2 · records	OracleSet tx A9AC5D56AFD8: Oracle C99DB849 (modified on the network): PreviousFields.PriceDataSeries ours= [{"PriceData":{"AssetPr…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-39	07:35 UTC	#107,645,890	Step 2 · records	OfferCreate tx 4AF7CE9301DC: DirectoryNode 75A6F1F7 (deleted on the network): FinalFields.PreviousTxnID ours="B24FBCEF8A81B0B60445BF0263D…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line, Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)
R-40	07:35 UTC	#107,645,891	Step 2 · records	Payment tx 1C31087B2CD9: RippleState DDB941EC (modified on the network): PreviousTxnID ours=null theirs="04C822468806C487C4467944BAF…	fixed: Step 2 record fixes (Batch marks, Oracle prices, AccountDelete balance, LP line,

ID	Time	Ledger(s)	Step	Description	Disposition
					Batch create-then-delete, directory links), live since 10-14 10:23 (cycle 173)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-10-14-01, INC-01 is INC-2026-10-14-01.

517 more receipts (#107,645,928–#107,662,483) are in the record and not listed individually; their dispositions: 517 fixed.

Incidents

INC-01 · 8 validations of a ledger hash the network did not validate

Window	2026-10-14 05:07 UTC to 2026-10-15 02:11 UTC
Ledgers	#107,643,613, #107,647,135, #107,648,513, #107,648,572, #107,648,595, #107,649,557, #107,655,959, #107,663,106
Impact	the validator signed a hash the network did not validate for these ledgers; data.xrpl.org scores such ledgers missed
Cause	Before cycle 183 the validator signed the ledger hash from the first peer that announced a ledger as accepted, without confirmation. For these ledgers that peer had built a different ledger, so the validator signed a hash the network never validated.
Action	commit d540e6e — fix(validator): sign a ledger only once the trusted validators reach quorum on its hash; deployed in cycle 183 at 2026-10-18 22:18 UTC

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
172	`493bc07`	06:43:56 UTC	2,063 passed / 0 failed; port 1,006 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches, HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	406 matched, 0 mismatched	#47 from 07:10 UTC: ended early: superseded by the next deployment (2026-10-14 14:23 UTC)
173	`850aec2`	14:23:15 UTC	2,112 passed / 0 failed; port 1,006 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	406 matched, 0 mismatched	#48 from 14:49 UTC: ended early: superseded by the next deployment (2026-10-14 16:41 UTC)
174	`534971d`	16:41:16 UTC	2,114 passed / 0 failed; port 1,006 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	408 matched, 0 mismatched	#49 from 17:07 UTC: ended early: superseded by the next deployment (2026-10-14 21:22 UTC)
175	`56f2653`	21:22:48 UTC	2,116 passed / 0 failed; port 1,006 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	409 matched, 0 mismatched	#50 from 21:49 UTC: ended early: superseded by the next deployment (2026-10-15 03:17 UTC)
176	`7684b00`	03:17:28 UTC	2,120 passed / 0 failed;	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	410 matched, 0 mismatched	#51 from 03:43 UTC: ended early: superseded by the next

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
			port 1,006 / 0			deployment (2026-10-15 14:13 UTC)

Cycle 172 delivered: `2524da0` feat(ledger): independence 2.1 — the field metadata table, generated from rippled 3.4.1; `48a92bf` feat(scripts): independence 2.1's check against mainnet's records, and an s2 answer-key fetcher; `524feca` fix(scripts): the 2.1 check reads only mainnet's l*_expected.json fixtures; `6564950` feat(ledger): independence 2.2 + 2.4 — each changed object's before and after, its threading, and the delivered amount; `2f5546f` feat(ledger): independence 2.3 — the AffectedNodes builder; `b2fa569` test(ledger): 2.3's canonical-form test borrows instead of cloning (clippy); `a2bf038` feat(probe): independence 2.6 — DX_META compares our metadata with mainnet's, byte for byte; `0257b15` fix(meta_check): TransactionResult goes to the codec as its number; `d4e2ca9` test(ledger): name the row type in 2.2's node_changes test (clippy); `8e50f5d` feat(ledger,node,probe): independence 2.4 Batch — the outer's own metadata and each applied inner's, ParentBatchID; `56c2acf` feat(ledger): independence 3.1 — rippled's canonical order, salted with the set's hash; `e1ce268` feat(probe): TXTREE — the ledger's transaction tree from our own records, against the header; `fa95e63` test(ledger): 3.1's mainnet check reads UNLModify's empty Account as the zero account; `8150b47` test(ledger): 3.1's mainnet check totals the runs and the results of later passes, for 3.2; `ccb673e` feat(node): independence 2.7 — the live shadow leg builds every record and checks our transaction tree against the header (XRPL_NATIVE_META=1); `c1fa280` test(node): counters_accumulate tolerates the parallel tests sharing the process-wide counters; `c5aeb2d` feat(ledger): independence 3.2(a) — rippled's apply passes, and an offline model of them; `6b0bf25` fix(ledger): 3.2(a) — a tecEXPIRED/tecKILLED from preclaim waits under tapRETRY like any tec; `4c54dce` Merge branch 'passes-3.2' into meta-2.2; `b63b2fd` feat(probe): independence 3.2(b) — DX_PASSES rebuilds the ledger from its set: canonical order, apply passes, the engine's own results; `68cc5b0` feat(scripts): fetch_meta_ledgers.py --parents, the header before each stored ledger (task 3.4); `f36632a` fix(probe): run_passes takes the maps, not borrowing closures; `4d8231d` feat(ledger): independence 3.4 — total_coins is the parent's less every fee charged; `76e622e` fix(ledger): a trust line is deleted as updated, so its DeletedNode FinalFields carry the new balance and flags (Mainnet #107541106 F3DC8E3560FD); `8a5973b` feat(ledger): independence 3.5 — pseudo-transactions get Change::preflight/preclaim's checks; `55b7b88` feat(scripts): fetch_meta_ledgers.py --range, a run of consecutive headers (task 5.3); `d67b170` feat(ledger): independence 5.3 — rippled's close time: resolution schedule, rounding, the no-consensus flag, the vote; `bf9999c` fix(ledger): 5.3's cadence checks use is_multiple_of (clippy's deny-by-default modulo_one); `643f8ea` docs: independence status 12 Oct — Step 2 built, 3.1/3.2/3.4/3.5 done offline, 5.3 in progress; `698bca3` docs(consensus): independence 5.1 — audit of the March consensus code against rippled 3.4.1; `473d2e4` Merge branch 'closetime-5.3' into meta-2.2; `152703f` Merge branch 'audit-5.1' into meta-2.2; `9be3a5b` fix(probe): DX_META and DX_PASSES give the engine the transaction's hash, which the fixture's tx_json lacks and the AccountTxnID stamp reads; `731395f` feat(node): independence 5.2(a) — shadow recorder for consensus rounds; `677f7e1` docs(node): 5.2(a) recorder — the measured disk figures; `c976229` Merge branch 'record-5.2a' into meta-2.2; `49aa2fe` fix(ledger): a payer short of the fee is tecINSUFF_FEE, charged what it holds; `5847f2c` feat(node): the validation quorum, pure, for 5.2(b); `f65d35d` fix(ledger): a missing account is terNO_ACCOUNT, a retry, not "tefNO_ACCOUNT"; `5d5e4b5` fix(ledger): a line TrustSet deletes is erased without its reserve bits, as mainnet records it (Mainnet #107541200); `beff740` fix(node): the amendment guard's /metrics lines start at the margin (the first build's carried the source indentation); `b83b414` feat(tools): xrpl-watch's last-event line shows its age, and FIXED with the finding once its cause has shipped (receipts_fixed.json beside the writer log); `5718682` docs: README rewritten for October 2026 — the native Rust engine writes the ledger state, what is still borrowed, the road to independence; `493bc07` Merge meta-2.2: independence Steps 2, 3 and 5 groundwork (own transaction records, canonical order and passes, fees, close time, the consensus round recorder) for cycle 172

Cycle 173 delivered: `a20a030` feat(node): independence 3.3 — the writer's own order and passes, in shadow (XRPL_NATIVE_ORDER=shadow|on, off by default); `6451e2a` feat(node): 3.3's shadow checks total XRP live (task 3.4): the parent's total less the fees our walk burned, each the Fee capped at what its payer held, against the header's; `1150110` test(node): differential_probe DX_SHADOW= — the live writer's propose() over a mainnet ledger's pre-state, in 3.3's shadow and on modes; `a02ac0a` test(node): DX_SHADOW builds the metadata key sets as the validator's ledger feed does (extract_affected_nodes, in TransactionIndex order); `4774af3` test(node): the 3.3 NoAccount audit's fixes, failing first; `7176e7c` fix(ledger): EscrowCancel refuses before CancelAfter and without one, whoever signs (A4); `881a780` fix(ledger): PermissionedDomainSet refuses a missing credential issuer, tecNO_ISSUER (A5); `22c2cf9` fix(ledger): NFTokenMint with Amount runs NFTokenCreateOffer's offer checks (A6); `dcee3e2` fix(ledger): Clawback tests that the holder exists before the issuer's flags; `9e3182f` fix(ledger): AMMBid requires every AuthAccounts account to exist, terNO_ACCOUNT (A3); `d1ffc5c` fix(node): checkPermission for a standalone delegated transaction (A2); `e7d153b` fix(ledger): a missing AMM pool is terNO_AMM on a complete state (A1); `87e400c` test(ledger): two EscrowCancel tests now cancel past CancelAfter, as rippled requires; `5a44553` fix(node): A2 refuses a missing Delegate object only on a complete state; `ecc2ac1` Merge order-3.3: independence 3.3, the writer's own order and passes (XRPL_NATIVE_ORDER, off by default), total XRP live, DX_SHADOW; `34cee5d` Merge noacct-fixes (5b): the 3.3 NoAccount audit's fixes — EscrowCancel's time gate, PermissionedDomainSet's issuer, NFTokenMint's offer checks, Clawback's order, AMMBid's AuthAccounts, checkPermission, terNO_AMM on a complete state; `3d9f510` feat(node): independence 4.1 — fetch a ledger from peers, proven to its ledger hash (shamap::wire, peer_fetch, peer_fetch_probe); `2343ee5` fix(node): peer_fetch's result takes the header before the closure; Hash256 for the tests only; `5dd3455` fix(node): peer_fetch_probe installs rustls' ring provider, as validation.rs does; `76670ef` feat(node): 4.1's live driver, off by default (XRPL_PEER_FETCH=1); `9ac9d82` feat(tools): xrpl-watch's INDEPENDENCE page (key i; 1 or Tab back): the road to a fully independent validator, live; `fa66644` feat(node): independence 4.5 — "validated" from our own count of trusted validations, in shadow (XRPL_OWN_VALIDATED=1, off by default); `74287b2` fix(node): 4.5 judges nothing before the validator list loads or for a ledger voted on before the tally listened; the TUI's step 4 shows our own "validated"; `4f2e38e` feat(node): independence 5.5, first form — our own ledger hash, trailing the network (XRPL_OWN_HASH=1, off by default); `57068cd` fix(tools): differential_probe's own-hash line reads rpc()'s answer as the result it already is; `092ac3b` docs(independence): 5.5's first form — our own ledger hash, trailing the network (ships in cycle 173, James #49); `9bd3693` fix(meta): UInt64 fields compare by value; AccountDelete and the LP-token line delete the zeroed SLE (Step 2 record receipts, soak #47); `c15899a` fix(meta): a Batch inner's record carries the earlier inners' threading for objects they created or it deletes (Step 2 receipts, soak #47); `850aec2` fix(meta): a Batch inner deleting what an earlier inner created keeps its final content; a deleted root page drops its links first (Step 2 receipts, soak #47)

Cycle 174 delivered: `01d5da5` fix(order): salt our canonical order with the set a quorum of trusted validations accepted (sfConsensusHash), as rippled does; the ledger's own root only without one (3.3 shadow receipts #107652744/745/752); `534971d` test(order): the own-root salt computed before the sort borrows the set

Cycle 175 delivered: `8cbf17f` fix(ffi): a Batch's expected mutation set folds in ledger order: a key one inner deleted and a later inner created again is a Modified, as libxrpl reports it (mutation_divergences #107636230 ×129, #107654753); `de5fd94` fix(payment): an existing destination is written back as rippled's update(sleDst) marks it, so a threadOwners stamp before it gives its FinalFields (Step 2 receipt #107655399 3B2D251B5861); `dfda531` fix(escrow): only EscrowFinish marks the destination's root written (rippled's update(sled)); a cancel leaves it threading-only (Step 2 receipt #107656021 1458D55C06B4); `eb0c569` fix(account): AccountDelete lowers the deleted account's OwnerCount (and TicketCount) per removed object, as each nonObligationDeleter does, and no longer writes an NFT offer's

Destination root back (update() audit); `56f2653` fix(offer): line_adjust's default-state test reads only the sender's QualityIn/QualityOut, as rippleCreditIOU does (update() audit, outside the class)

Cycle 176 delivered: `a366749` fix(amm): a withdrawal or clawback that empties the pool erases the AMM with the balance it had: deleteAMMAccountIfEmpty writes the new LPTokenBalance only when the AMM survives (Step 2 receipt #107662483 88C9F7289CD1); `7a4bd78` fix(peers): a peer that hides its version is "hidden", not its first peer's version; the node names itself xrpl-rust-validator/0.4.0+; `7684b00` Merge peer-version (a team session, James's ask): hidden peer versions; the node's version xrpl-rust-validator/0.4.0+

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
11:05 UTC	xrpl-sdk	`c1a42c1`	fix(codec): an AccountID field never set is an empty VL, "" in JSON
11:06 UTC	xrpl-validator	`092ac3b`	docs(independence): 5.5's first form — our own ledger hash, trailing the network (ships in cycle 173, James #49)
11:06 UTC	xrpl-validator	`57068cd`	fix(tools): differential_probe's own-hash line reads rpc()'s answer as the result it already is
11:06 UTC	xrpl-validator	`4f2e38e`	feat(node): independence 5.5, first form — our own ledger hash, trailing the network (XRPL_OWN_HASH=1, off by default)
11:06 UTC	xrpl-validator	`74287b2`	fix(node): 4.5 judges nothing before the validator list loads or for a ledger voted on before the tally listened; the TUI's step 4 shows our own "validated"
11:06 UTC	xrpl-validator	`fa66644`	feat(node): independence 4.5 — "validated" from our own count of trusted validations, in shadow (XRPL_OWN_VALIDATED=1, off by default)
11:06 UTC	xrpl-validator	`9ac9d82`	feat(tools): xrpl-watch's INDEPENDENCE page (key i; 1 or Tab back): the road to a fully independent validator, live
11:06 UTC	xrpl-validator	`76670ef`	feat(node): 4.1's live driver, off by default (XRPL_PEER_FETCH=1)
11:06 UTC	xrpl-validator	`5dd3455`	fix(node): peer_fetch_probe installs rustls' ring provider, as validation.rs does
11:06 UTC	xrpl-validator	`2343ee5`	fix(node): peer_fetch's result takes the header before the closure; Hash256 for the tests only
11:06 UTC	xrpl-validator	`3d9f510`	feat(node): independence 4.1 — fetch a ledger from peers, proven to its ledger hash (shamap::wire, peer_fetch, peer_fetch_probe)
11:06 UTC	xrpl-validator	`34cee5d`	Merge noacct-fixes (5b): the 3.3 NoAccount audit's fixes — EscrowCancel's time gate, PermissionedDomainSet's issuer, NFTokenMint's offer checks, Clawback's order, AMMBid's AuthAccounts, checkPermission, terNO_AMM on a complete state
11:06 UTC	xrpl-validator	`ecc2ac1`	Merge order-3.3: independence 3.3, the writer's own order and passes (XRPL_NATIVE_ORDER, off by default), total XRP live, DX_SHADOW
11:06 UTC	xrpl-validator	`5a44553`	fix(node): A2 refuses a missing Delegate object only on a complete state
11:06 UTC	xrpl-validator	`87e400c`	test(ledger): two EscrowCancel tests now cancel past CancelAfter, as rippled requires
11:06 UTC	xrpl-validator	`e7d153b`	fix(ledger): a missing AMM pool is terNO_AMM on a complete state (A1)
11:06 UTC	xrpl-validator	`d1ffc5c`	fix(node): checkPermission for a standalone delegated transaction (A2)
11:06 UTC	xrpl-validator	`9e3182f`	fix(ledger): AMMBid requires every AuthAccounts account to exist, terNO_ACCOUNT (A3)

Merged	Repo	Commit	Subject
11:06 UTC	xrpl-validator	`dcee3e2`	fix(ledger): Clawback tests that the holder exists before the issuer's flags
11:06 UTC	xrpl-validator	`22c2cf9`	fix(ledger): NFTokenMint with Amount runs NFTokenCreateOffer's offer checks (A6)
11:06 UTC	xrpl-validator	`881a780`	fix(ledger): PermissionedDomainSet refuses a missing credential issuer, tecNO_ISSUER (A5)
11:06 UTC	xrpl-validator	`7176e7c`	fix(ledger): EscrowCancel refuses before CancelAfter and without one, whoever signs (A4)
11:06 UTC	xrpl-validator	`4774af3`	test(node): the 3.3 NoAccount audit's fixes, failing first
11:06 UTC	xrpl-validator	`a02ac0a`	test(node): DX_SHADOW builds the metadata key sets as the validator's ledger feed does (extract_affected_nodes, in TransactionIndex order)
11:06 UTC	xrpl-validator	`1150110`	test(node): differential_probe DX_SHADOW= — the live writer's propose() over a mainnet ledger's pre-state, in 3.3's shadow and on modes
11:06 UTC	xrpl-validator	`6451e2a`	feat(node): 3.3's shadow checks total XRP live (task 3.4): the parent's total less the fees our walk burned, each the Fee capped at what its payer held, against the header's
11:06 UTC	xrpl-validator	`a20a030`	feat(node): independence 3.3 — the writer's own order and passes, in shadow (XRPL_NATIVE_ORDER=shadow|on, off by default)
13:31 UTC	xrpl-validator	`9bd3693`	fix(meta): UInt64 fields compare by value; AccountDelete and the LP-token line delete the zeroed SLE (Step 2 record receipts, soak #47)
13:31 UTC	xrpl-validator	`c15899a`	fix(meta): a Batch inner's record carries the earlier inners' threading for objects they created or it deletes (Step 2 receipts, soak #47)
13:31 UTC	xrpl-validator	`850aec2`	fix(meta): a Batch inner deleting what an earlier inner created keeps its final content; a deleted root page drops its links first (Step 2 receipts, soak #47)
15:16 UTC	xrpl-validator	`01d5da5`	fix(order): salt our canonical order with the set a quorum of trusted validations accepted (sfConsensusHash), as rippled does; the ledger's own root only without one (3.3 shadow receipts #107652744/745/752)
15:16 UTC	xrpl-validator	`534971d`	test(order): the own-root salt computed before the sort borrows the set
17:34 UTC	xrpl-validator	`8cbf17f`	fix(ffi): a Batch's expected mutation set folds in ledger order: a key one inner deleted and a later inner created again is a Modified, as libxrpl reports it (mutation_divergences #107636230 ×129, #107654753)
19:10 UTC	xrpl-validator	`de5fd94`	fix(payment): an existing destination is written back as rippled's update(sleDst) marks it, so a threadOwners stamp before it gives its FinalFields (Step 2 receipt #107655399 3B2D251B5861)
19:10 UTC	xrpl-validator	`dfda531`	fix(escrow): only EscrowFinish marks the destination's root written (rippled's update(sled)); a cancel leaves it threading-only (Step 2 receipt #107656021 1458D55C06B4)
19:10 UTC	xrpl-validator	`eb0c569`	fix(account): AccountDelete lowers the deleted account's OwnerCount (and TicketCount) per removed object, as each nonObligationDeleter does, and no longer writes an NFT offer's Destination root back (update() audit)
19:10 UTC	xrpl-validator	`56f2653`	fix(offer): line_adjust's default-state test reads only the sender's QualityIn/QualityOut, as rippleCreditIOU does (update() audit, outside the class)
02:23 UTC	xrpl-validator	`a366749`	fix(amm): a withdrawal or clawback that empties the pool erases the AMM with the balance it had: deleteAMMAccountIfEmpty writes the new LPTokenBalance only when the AMM survives (Step 2 receipt #107662483 88C9F7289CD1)
02:23 UTC	xrpl-validator	`7a4bd78`	fix(peers): a peer that hides its version is "hidden", not its first peer's version; the node names itself xrpl-rust-validator/0.4.0+
02:23 UTC	xrpl-validator	`7684b00`	Merge peer-version (a team session, James's ask): hidden peer versions; the node's version xrpl-rust-validator/0.4.0+

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(codec): an AccountID field never set is an empty VL, "" in JSON** (`c1a42c1`, xrpl-sdk) rippled serializes a default STAccount (never set) as an empty VL and shows it as "" in JSON: a UNLModify pseudo-transaction's Account is `81 00` (mainnet #107068416), while an EnableAmendment sets its Account to the zero account, `81 14` + 20 zeros (rrrrrrrrrrrrrrrrrrrrrhoLvTp, #106911489). The decoder rejected any length but 20, so a UNLModify blob would not decode, and the encoder could not write "" at all.
- **fix(tools): differential_probe's own-hash line reads rpc()'s answer as the result it already is** (`57068cd`, xrpl-validator)
- **feat(node): independence 5.5, first form — our own ledger hash, trailing the network (XRPL_OWN_HASH=1, off by default)** (`4f2e38e`, xrpl-validator) own_hash.rs assembles each landed ledger's header from what this validator made itself and hashes it (calculateLedgerHash): the state root from our hasher (ours when every byte written was our engine's, the skip list and protocol singletons included), the transaction root from our own records (ours when our order matched, 3.3), the total XRP as our previous total less our burn, the parent hash, resolution (5.3) and …
- **fix(node): 4.5 judges nothing before the validator list loads or for a ledger voted on before the tally listened; the TUI's step 4 shows our own "validated"** (`74287b2`, xrpl-validator) The first mainnet run's one short ledger was the one closing as the sessions connected.
- **feat(node): independence 4.5 — "validated" from our own count of trusted validations, in shadow (XRPL_OWN_VALIDATED=1, off by default)** (`fa66644`, xrpl-validator) own_validated counts, ledger by ledger, the signature-checked full validations from validators on the verified list (signing key to master through the list's manifests), less the negative UNL the newest written ledger carries, against rippled's quorum (consensus::quorum: max(ceil(0.8 x effective), ceil(0.6 x UNL))), as LedgerMaster::checkAccept does. Every ledger the validator's ledger feed takes because the reference node calls it validated is …
- **feat(tools): xrpl-watch's INDEPENDENCE page (key i; 1 or Tab back): the road to a fully independent validator, live** (`9ac9d82`, xrpl-validator) The six steps of the plan as a road (LIVE / SHADOW / BUILT · OFF / STARTED / LATER), each with the running validator's own numbers: the amendment guard, signature checks and their enforcement, the validator list and whether it fails closed; our own records and trees (native_meta); our own order, results, bytes and total XRP (native_order); ledgers from peers (peer_fetch); the consensus round recorder and the signing …
- **feat(node): 4.1's live driver, off by default (XRPL_PEER_FETCH=1)** (`76670ef`, xrpl-validator) Each validated ledger the validator's ledger feed has proven against its header's transaction root (the tx-gate's pairs) is fetched from peers too and compared, byte for byte: peer_fetch::offer_network. The driver works one ledger at a time and spreads its requests: each peer at most once every 8 s, the longest-rested first, another one after a 6 s silence, and a peer that misses three replies (or sends a header that is not the …
- **fix(node): peer_fetch_probe installs rustls' ring provider, as validation.rs does** (`5dd3455`, xrpl-validator)
- **fix(node): peer_fetch's result takes the header before the closure; Hash256 for the tests only** (`2343ee5`, xrpl-validator)
- **feat(node): independence 4.1 — fetch a ledger from peers, proven to its ledger hash (shamap::wire, peer_fetch, peer_fetch_probe)** (`3d9f510`, xrpl-validator) shamap::wire reads rippled's SHAMap wire format (serializeForWire, SHAMapNodeID::getRawString) and assembles a ledger's transaction tree from fetched nodes,

taking a node only at a position an accepted parent named and only when it hashes to what the parent holds there. peer_fetch::LedgerFetch drives rippled's own requests (liBASE for the header and the tree's root, then liTX_NODE three levels at a time) and …

- **fix(node): A2 refuses a missing Delegate object only on a complete state** (`5a44553`, xrpl-validator) Live-path change.
- **fix(ledger): a missing AMM pool is terNO_AMM on a complete state (A1)** (`e7d153b`, xrpl-validator) Live-path change. It only bites when we build a ledger ourselves.
- **fix(node): checkPermission for a standalone delegated transaction (A2)** (`d1ffc5c`, xrpl-validator) Live-path change. It only bites when we build a ledger ourselves.
- **fix(ledger): AMMBid requires every AuthAccounts account to exist, terNO_ACCOUNT (A3)** (`9e3182f`, xrpl-validator) Live-path change. It only bites when we build a ledger ourselves.
- **fix(ledger): Clawback tests that the holder exists before the issuer's flags** (`dcee3e2`, xrpl-validator) Live-path change. It only bites when we build a ledger ourselves.
- **fix(ledger): NFTokenMint with Amount runs NFTokenCreateOffer's offer checks (A6)** (`22c2cf9`, xrpl-validator) Live-path change.
- **fix(ledger): PermissionedDomainSet refuses a missing credential issuer, tecNO_ISSUER (A5)** (`881a780`, xrpl-validator) Live-path change.
- **fix(ledger): EscrowCancel refuses before CancelAfter and without one, whoever signs (A4)** (`7176e7c`, xrpl-validator) Live-path change.
- **feat(node): 3.3's shadow checks total XRP live (task 3.4): the parent's total less the fees our walk burned, each the Fee capped at what its payer held, against the header's** (`6451e2a`, xrpl-validator) A receipt and a counter, never a refusal: the total is in the header, which the writer does not write.
- **feat(node): independence 3.3 — the writer's own order and passes, in shadow (XRPL_NATIVE_ORDER=shadow|on, off by default)** (`a20a030`, xrpl-validator) Shadow applies each ledger in rippled's canonical order and retry passes first, keeps the order, the results and the bytes that gives, undoes it, and applies in the network's order as before; native_order counts and logs where the two differ (/api/engine native_order, /metrics, native_order.jsonl). On makes our order the apply, and a ledger it cannot reproduce a writer refusal. The per-transaction body moves into …
- **fix(meta): UInt64 fields compare by value; AccountDelete and the LP-token line delete the zeroed SLE (Step 2 record receipts, soak #47)** (`9bd3693`, xrpl-validator)
- **fix(meta): a Batch inner's record carries the earlier inners' threading for objects they created or it deletes (Step 2 receipts, soak #47)** (`c15899a`, xrpl-validator)
- **fix(meta): a Batch inner deleting what an earlier inner created keeps its final content; a deleted root page drops its links first (Step 2 receipts, soak #47)** (`850aec2`, xrpl-validator)
- **fix(order): salt our canonical order with the set a quorum of trusted validations accepted (sfConsensusHash), as rippled does; the ledger's own root only without one (3.3 shadow receipts #107652744/745/752)** (`01d5da5`, xrpl-validator)
- **fix(ffi): a Batch's expected mutation set folds in ledger order: a key one inner deleted and a later inner created again is a Modified, as libxrpl reports it (mutation_divergences #107636230 ×129, #107654753)** (`8cbf17f`, xrpl-validator)
- **fix(payment): an existing destination is written back as rippled's update(sleDst) marks it, so a threadOwners stamp before it gives its FinalFields (Step 2 receipt #107655399 3B2D251B5861)** (`de5fd94`,

xrpl-validator)

- **fix(escrow): only EscrowFinish marks the destination's root written (rippled's update(sled)); a cancel leaves it threading-only (Step 2 receipt #107656021 1458D55C06B4)** (`dfda531`, xrpl-validator)
- **fix(account): AccountDelete lowers the deleted account's OwnerCount (and TicketCount) per removed object, as each nonObligationDeleter does, and no longer writes an NFT offer's Destination root back (update() audit)** (`eb0c569`, xrpl-validator)
- **fix(offer): line_adjust's default-state test reads only the sender's QualityIn/QualityOut, as rippleCreditIOU does (update() audit, outside the class)** (`56f2653`, xrpl-validator)
- **fix(amm): a withdrawal or clawback that empties the pool erases the AMM with the balance it had: deleteAMMAccountIfEmpty writes the new LPTokenBalance only when the AMM survives (Step 2 receipt #107662483 88C9F7289CD1)** (`a366749`, xrpl-validator)
- **fix(peers): a peer that hides its version is "hidden", not its first peer's version; the node names itself xrpl-rust-validator/0.4.0+** (`7a4bd78`, xrpl-validator) The peer page's crawler fell back to overlay.active[0].version when a node's /crawl had no server section. The three XRP Kuwait hubs (hubs.xrpkuwait.com) hide theirs and list us first, so they showed as "xrpl-rust-validator/0.1.0", our own name. version::crawl_label now keeps rippled numbers ("rippled-3.4.1"), keeps named builds as they are, and says "hidden" otherwise.

Tests and campaigns

- **test(ledger): two EscrowCancel tests now cancel past CancelAfter, as rippled requires** (`87e400c`, xrpl-validator)
- **test(node): the 3.3 NoAccount audit's fixes, failing first** (`4774af3`, xrpl-validator)
- **test(node): DX_SHADOW builds the metadata key sets as the validator's ledger feed does (extract_affected_nodes, in TransactionIndex order)** (`a02ac0a`, xrpl-validator)
- **test(node): differential_probe DX_SHADOW= — the live writer's propose() over a mainnet ledger's pre-state, in 3.3's shadow and on modes** (`1150110`, xrpl-validator)
- **test(order): the own-root salt computed before the sort borrows the set** (`534971d`, xrpl-validator)

Docs and operations

- **docs(independence): 5.5's first form — our own ledger hash, trailing the network (ships in cycle 173, James #49)** (`092ac3b`, xrpl-validator)
- **Merge noacct-fixes (5b): the 3.3 NoAccount audit's fixes — EscrowCancel's time gate, PermissionedDomainSet's issuer, NFTokenMint's offer checks, Clawback's order, AMMBid's AuthAccounts, checkPermission, terNO_AMM on a complete state** (`34cee5d`, xrpl-validator)
- **Merge order-3.3: independence 3.3, the writer's own order and passes (XRPL_NATIVE_ORDER, off by default), total XRP live, DX_SHADOW** (`ecc2ac1`, xrpl-validator)
- **Merge peer-version (a team session, James's ask): hidden peer versions; the node's version xrpl-rust-validator/0.4.0+** (`7684b00`, xrpl-validator)

Independence program

No independence record for UTC day 2026-10-14; records begin 2026-10-20.

Network

Measure	Value
Ledgers closed	22,202 (#107,642,576–#107,664,777)
Mean close interval	3.89 s
Longest close interval	10 s (before #107,663,064)
Closes at least 15 s apart	0
Empty ledgers	0

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_4_0	2026-10-09 09:34 UTC	2026-10-23 09:34 UTC

Amendments enabled on mainnet: 97.

Open items

- Amendment fixCleanup3_4_0: activation due ~2026-10-23 09:34 UTC.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
10-12 16:32	10-14 11:05	xrpl-sdk	`c1a42c1`	fix(codec): an AccountID field never set is an empty VL, "" in JSON
10-14 02:13	10-14 11:06	xrpl-validator	`092ac3b`	docs(independence): 5.5's first form — our own ledger hash, trailing the network (ships in cycle 173, James #49)
10-14 01:50	10-14 11:06	xrpl-validator	`57068cd`	fix(tools): differential_probe's own-hash line reads rpc()'s answer as the result it already is
10-14 01:42	10-14 11:06	xrpl-validator	`4f2e38e`	feat(node): independence 5.5, first form — our own ledger hash, trailing the network (XRPL_OWN_HASH=1, off by default)
10-14 00:52	10-14 11:06	xrpl-validator	`74287b2`	fix(node): 4.5 judges nothing before the validator list loads or for a ledger voted on before the tally listened; the TUI's step 4 shows our own "validated"

Committed	Merged into main	Repo	Commit	Subject
10-14 00:41	10-14 11:06	xrpl-validator	`fa66644`	feat(node): independence 4.5 — "validated" from our own count of trusted validations, in shadow (XRPL_OWN_VALIDATED=1, off by default)
10-13 21:52	10-14 11:06	xrpl-validator	`9ac9d82`	feat(tools): xrpl-watch's INDEPENDENCE page (key i; 1 or Tab back): the road to a fully independent validator, live
10-13 21:34	10-14 11:06	xrpl-validator	`76670ef`	feat(node): 4.1's live driver, off by default (XRPL_PEER_FETCH=1)
10-13 21:28	10-14 11:06	xrpl-validator	`5dd3455`	fix(node): peer_fetch_probe installs rustls' ring provider, as validation.rs does
10-13 21:27	10-14 11:06	xrpl-validator	`2343ee5`	fix(node): peer_fetch's result takes the header before the closure; Hash256 for the tests only
10-13 21:26	10-14 11:06	xrpl-validator	`3d9f510`	feat(node): independence 4.1 — fetch a ledger from peers, proven to its ledger hash (shamap::wire, peer_fetch, peer_fetch_probe)
10-13 20:41	10-14 11:06	xrpl-validator	`34cee5d`	Merge noacct-fixes (5b): the 3.3 NoAccount audit's fixes — EscrowCancel's time gate, PermissionedDomainSet's issuer, NFTokenMint's offer checks, Clawback's order, AMMBid's AuthAccounts, checkPermission, terNO_AMM on a complete state
10-13 20:41	10-14 11:06	xrpl-validator	`ecc2ac1`	Merge order-3.3: independence 3.3, the writer's own order and passes (XRPL_NATIVE_ORDER, off by default), total XRP live, DX_SHADOW
10-13 20:31	10-14 11:06	xrpl-validator	`5a44553`	fix(node): A2 refuses a missing Delegate object only on a complete state
10-13 20:31	10-14 11:06	xrpl-validator	`87e400c`	test(ledger): two EscrowCancel tests now cancel past CancelAfter, as rippled requires
10-13 20:18	10-14 11:06	xrpl-validator	`e7d153b`	fix(ledger): a missing AMM pool is terNO_AMM on a complete state (A1)
10-13 20:15	10-14 11:06	xrpl-validator	`d1ffc5c`	fix(node): checkPermission for a standalone delegated transaction (A2)
10-13 20:15	10-14 11:06	xrpl-validator	`9e3182f`	fix(ledger): AMMBid requires every AuthAccounts account to exist, terNO_ACCOUNT (A3)
10-13 20:15	10-14 11:06	xrpl-validator	`dcee3e2`	fix(ledger): Clawback tests that the holder exists before the issuer's flags
10-13 20:15	10-14 11:06	xrpl-validator	`22c2cf9`	fix(ledger): NFTokenMint with Amount runs NFTokenCreateOffer's offer checks (A6)
10-13 20:15	10-14 11:06	xrpl-validator	`881a780`	fix(ledger): PermissionedDomainSet refuses a missing credential issuer, tecNO_ISSUER (A5)
10-13 20:14	10-14 11:06	xrpl-validator	`7176e7c`	fix(ledger): EscrowCancel refuses before CancelAfter and without one, whoever signs (A4)
10-13 20:14	10-14 11:06	xrpl-validator	`4774af3`	test(node): the 3.3 NoAccount audit's fixes, failing first
10-13 19:46	10-14 11:06	xrpl-validator	`a02ac0a`	test(node): DX_SHADOW builds the metadata key sets as the validator's ledger feed does (extract_affected_nodes, in TransactionIndex order)
10-13 19:33	10-14 11:06	xrpl-validator	`1150110`	test(node): differential_probe DX_SHADOW= — the live writer's propose() over a mainnet ledger's pre-state, in 3.3's shadow and on modes
10-13 19:31	10-14 11:06	xrpl-validator	`6451e2a`	feat(node): 3.3's shadow checks total XRP live (task 3.4): the parent's total less the fees our walk burned, each the Fee capped at what its payer held, against the header's
10-13 19:17	10-14 11:06	xrpl-validator	`a20a030`	feat(node): independence 3.3 — the writer's own order and passes, in shadow (XRPL_NATIVE_ORDER=shadow|on, off by default)
10-14 11:26	10-14 13:31	xrpl-validator	`9bd3693`	fix(meta): UInt64 fields compare by value; AccountDelete and the LP-token line delete the zeroed SLE (Step 2 record receipts, soak #47)
10-14 11:43	10-14 13:31	xrpl-validator	`c15899a`	fix(meta): a Batch inner's record carries the earlier inners' threading for objects they created or it deletes (Step 2 receipts, soak #47)
10-14 12:59	10-14 13:31	xrpl-validator	`850aec2`	fix(meta): a Batch inner deleting what an earlier inner created keeps its final content; a deleted root page drops its links first (Step 2 receipts, soak #47)

Committed	Merged into main	Repo	Commit	Subject
10-14 15:14	10-14 15:16	xrpl-validator	`01d5da5`	fix(order): salt our canonical order with the set a quorum of trusted validations accepted (sfConsensusHash), as rippled does; the ledger's own root only without one (3.3 shadow receipts #107652744/745/752)
10-14 15:15	10-14 15:16	xrpl-validator	`534971d`	test(order): the own-root salt computed before the sort borrows the set
10-14 17:15	10-14 17:34	xrpl-validator	`8cbf17f`	fix(ffi): a Batch's expected mutation set folds in ledger order: a key one inner deleted and a later inner created again is a Modified, as libxrpl reports it (mutation_divergences #107636230 ×129, #107654753)
10-14 18:08	10-14 19:10	xrpl-validator	`de5fd94`	fix(payment): an existing destination is written back as rippled's update(sleDst) marks it, so a threadOwners stamp before it gives its FinalFields (Step 2 receipt #107655399 3B2D251B5861)
10-14 18:37	10-14 19:10	xrpl-validator	`dfda531`	fix(escrow): only EscrowFinish marks the destination's root written (rippled's update(sled)); a cancel leaves it threading-only (Step 2 receipt #107656021 1458D55C06B4)
10-14 18:52	10-14 19:10	xrpl-validator	`eb0c569`	fix(account): AccountDelete lowers the deleted account's OwnerCount (and TicketCount) per removed object, as each nonObligationDeleter does, and no longer writes an NFT offer's Destination root back (update() audit)
10-14 18:54	10-14 19:10	xrpl-validator	`56f2653`	fix(offer): line_adjust's default-state test reads only the sender's QualityIn/QualityOut, as rippleCreditIOU does (update() audit, outside the class)
10-15 01:54	10-15 02:23	xrpl-validator	`a366749`	fix(amm): a withdrawal or clawback that empties the pool erases the AMM with the balance it had: deleteAMMAccountIfEmpty writes the new LPTokenBalance only when the AMM survives (Step 2 receipt #107662483 88C9F7289CD1)
10-15 02:20	10-15 02:23	xrpl-validator	`7a4bd78`	fix(peers): a peer that hides its version is "hidden", not its first peer's version; the node names itself xrpl-rust-validator/0.4.0+
10-15 02:21	10-15 02:23	xrpl-validator	`7684b00`	Merge peer-version (a team session, James's ask): hidden peer versions; the node's version xrpl-rust-validator/0.4.0+

Times in UTC. A commit appears on the day it was committed to main or merged into it.